SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS

WINTER SEMESTER 2025 - 2026

FINAL ASSESSMENT TEST

April 15, 2026

Programme Name : M.Tech Software Engineering (Integrated)

Course Mode: Lab Based Component

Class Number: VL2025260503869

Course: ISWE406P - Agile Development Process and Devops Lab

Faculty: Dr. N. BALA GANESH

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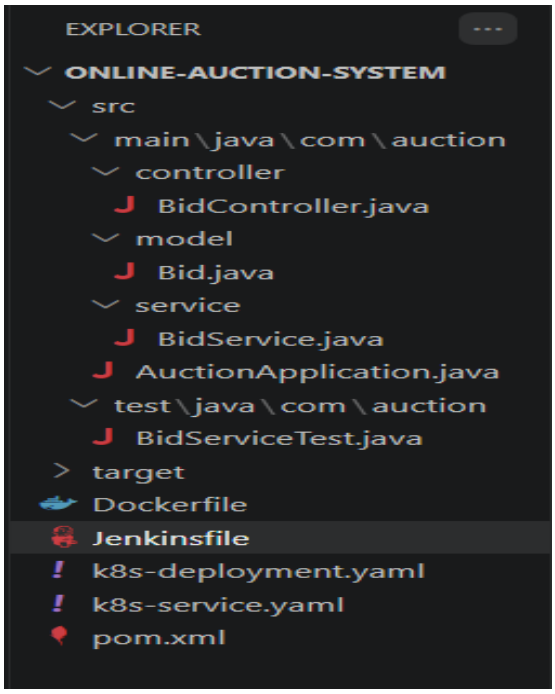
Reg no:-23MIS0433

Answer:-

i)Java program development:-

Folder structure:-



Java files:-

src > main > java > com > auction > AuctionApplication.java > AuctionApplication

```java
package com.auction;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class AuctionApplication {
    Run main | Debug main
    public static void main(String[] args) {
        SpringApplication.run(AuctionApplication.class, args);
    }
}
```

src > main > java > com > auction > model > Bid.java > Bid

```java
package com.auction.model;

public class Bid {
    private String bidder;
    private double amount;

    public Bid() {}

    public Bid(String bidder, double amount) {
        this.bidder = bidder;
        this.amount = amount;
    }

    public String getBidder() { return bidder; }
    public void setBidder(String bidder) { this.bidder = bidder; }

    public double getAmount() { return amount; }
    public void setAmount(double amount) { this.amount = amount; }
}
```

src > main > java > com > auction > controller > BidController.java > BidController

```java
package com.auction.controller;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;

import com.auction.model.Bid;
import com.auction.service.BidService;

@RestController
@RequestMapping("/bids")
public class BidController {

    @Autowired
    private BidService service;

    @PostMapping
    public String placeBid(@RequestBody Bid bid) {
        return service.placeBid(bid);
    }

    @GetMapping("/highest")
    public Bid getHighest() {
        return service.getHighestBid();
    }
}
```

src > main > java > com > auction > model > Bid.java > Bid

```java
package com.auction.model;

public class Bid {
    private String bidder;
    private double amount;

    public Bid() {}

    public Bid(String bidder, double amount) {
        this.bidder = bidder;
        this.amount = amount;
    }

    public String getBidder() { return bidder; }
    public void setBidder(String bidder) { this.bidder = bidder; }

    public double getAmount() { return amount; }
    public void setAmount(double amount) { this.amount = amount; }
}
```

ii)Junit file:-

src > test > java > com > auction > BidServiceTest.java > BidServiceTest

```java
package com.auction;

import static org.junit.jupiter.api.Assertions.assertEquals;
import org.junit.jupiter.api.Test;

import com.auction.model.Bid;
import com.auction.service.BidService;

public class BidServiceTest {

    @Test
    void validBid() {
        BidService service = new BidService();
        assertEquals("Bid accepted", service.placeBid(new Bid("Ajay", 200)));
    }

    @Test
    void lowBid() {
        BidService service = new BidService();
        service.placeBid(new Bid("A", 200));
        assertEquals("Bid too low", service.placeBid(new Bid("B", 100)));
    }

    @Test
    void invalidBid() {
        BidService service = new BidService();
        assertEquals("Invalid bid", service.placeBid(new Bid("A", -1)));
    }
}
```

k8s-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: auction-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: auction
  template:
    metadata:
      labels:
        app: auction
    spec:
      containers:
      - name: auction
        image: ajay222006/auction:latest
        ports:
        - containerPort: 8080
```

k8s-service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: auction-service
spec:
  type: NodePort
  selector:
    app: auction
  ports:
    - port: 80
      targetPort: 8080
      nodePort: 30007
```

pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0">
  <modelVersion>4.0.0</modelVersion>

  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.2.0</version>
  </parent>

  <groupId>com.auction</groupId>
  <artifactId>auction</artifactId>
  <version>1.0</version>

  <dependencies>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-test</artifactId>
      <scope>test</scope>
    </dependency>
  </dependencies>

  <build>
    <plugins>
      <plugin>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-maven-plugin</artifactId>
      </plugin>
    </plugins>
  </build>
</project>
```

Docker file:-

```
Dockerfile
FROM openjdk:17
COPY target/auction-1.0.jar app.jar
ENTRYPOINT ["java","-jar","/app.jar"]
```

Jenkins file:-

```
Jenkinsfile
pipeline {
    agent any

    environment {
        DOCKER_IMAGE = "ajay222006/auction"
    }

    stages {

        stage('Clone') {
            steps {
                git 'https://github.com/Ajay222006/Online-Auction-System.git'
            }
        }

        stage('Build') {
            steps {
                bat 'mvn clean package'
            }
        }

        stage('Test') {
            steps {
                bat 'mvn test'
            }
        }

        stage('Docker Build') {
            steps {
                bat 'docker build -t %DOCKER_IMAGE% .'
            }
        }

        stage('Docker Push') {
            steps {
                withCredentials([usernamePassword(credentialsId: 'dockerhub', usernameVariable: 'USER', passwordVariable: 'PASS')]) {
                    bat 'echo %PASS% | docker login -u %USER% --password-stdin'
                    bat 'docker push %DOCKER_IMAGE%'
                }
            }
        }

```

Maven build:-

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system> mvn clean package
Downloading from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.13.0/commons-io-2.13.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm-tree/9.5/asm-tree-9.5.jar (52 kB at 4.3 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/vafer/jdependency/2.8.0/jdependency-2.8.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-dependency-tree/3.2.1/maven-dependency-tree-3.2.1.jar (43 kB at
 3.5 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-collections4/4.4/commons-collections4-4.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/vafer/jdependency/2.8.0/jdependency-2.8.0.jar (233 kB at 17 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/jdom/jdom2/2.0.6.1/jdom2-2.0.6.1.jar (328 kB at 24 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/net/java/dev/jna/jna/5.13.0/jna-5.13.0.jar (1.9 MB at 133 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.13.0/commons-io-2.13.0.jar (484 kB at 34 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-collections4/4.4/commons-collections4-4.4.jar (752 kB at 46 kB/s)
[INFO] Replacing main artifact C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system\target\auction-1.0.jar with repackaged archive, adding nested
 dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system\target\auction-1.0.jar.original
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  44.451 s
[INFO] Finished at: 2026-04-15T18:09:08+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system>
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system> mvn test
[INFO]
[INFO] --- surefire:3.1.2:test (default-test) @ auction ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.auction.BidServiceTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.045 s -- in com.auction.BidServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.529 s
[INFO] Finished at: 2026-04-15T18:10:33+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system>
```

Git push:-

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system> git push -u origin main
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/TEST-com.auction.BidServiceTest.xml
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/TEST-com.auction.BidServiceTest.xml
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/TEST-com.auction.BidServiceTest.xml
 create mode 100644 target/surefire-reports/com.auction.BidServiceTest.txt
 create mode 100644 target/surefire-reports/com.auction.BidServiceTest.txt
 create mode 100644 target/test-classes/com/auction/BidServiceTest.class
PS C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system> git branch -M main
PS C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system> git remote add origin https://github.com/Ajay222006/Online-Auction-System.git
PS C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system> git push -u origin main
Enumerating objects: 56, done.
Enumerating objects: 56, done.
Counting objects: 100% (56/56), done.
Delta compression using up to 12 threads
Counting objects: 100% (56/56), done.
Delta compression using up to 12 threads
Compressing objects: 100% (33/33), done.
Delta compression using up to 12 threads
Compressing objects: 100% (33/33), done.
Compressing objects: 100% (33/33), done.
Writing objects: 100% (56/56), 16.88 MiB | 35.49 MiB/s, done.
Total 56 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Ajay222006/Online-Auction-System.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system>
```

iii)CI/CD pipeline:-

Pipeline script from SCM

SCM ?

Git ?

Repositories ?

Repository URL ?

https://github.com/Ajay222006/Online-Auction-System.git

Credentials ?

ajay222006/****** (Docker-push) + Add

Advanced

+ Add Repository

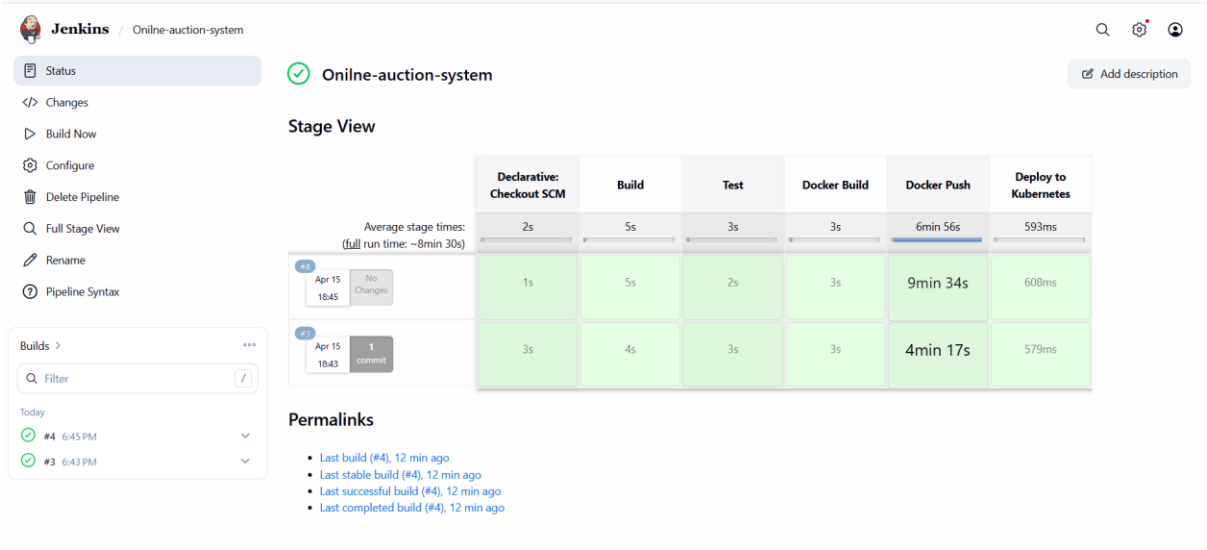
Branches to build ?

Branch Specifier (blank for 'any') ?

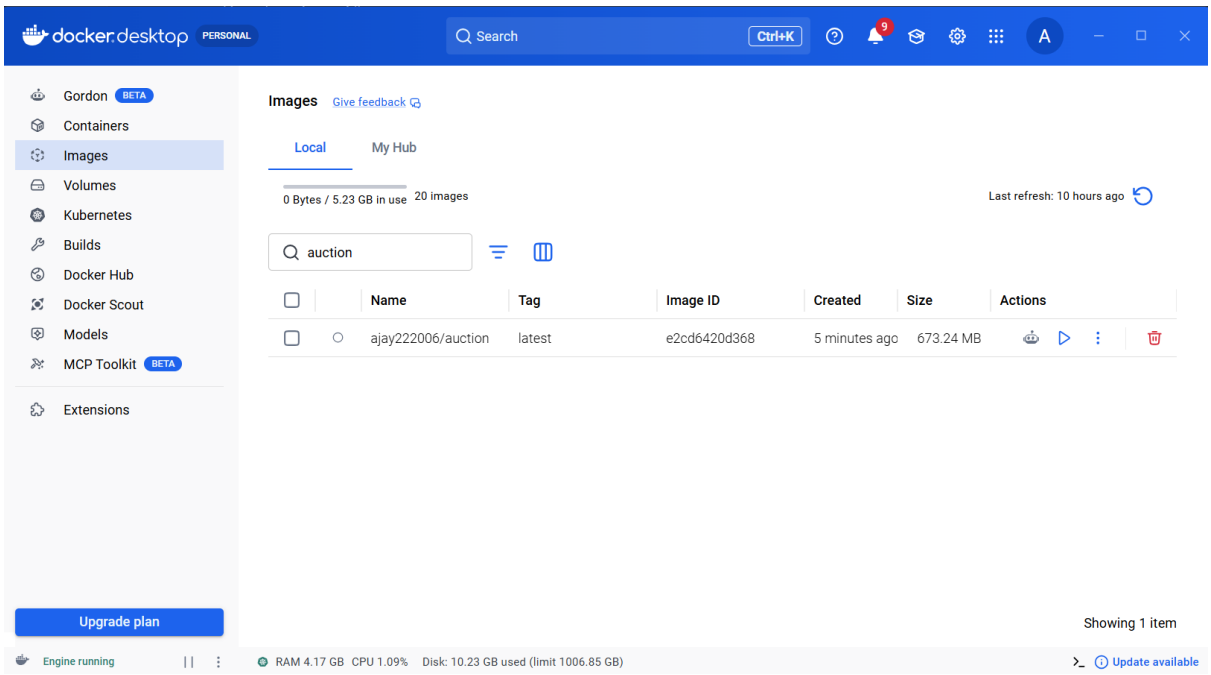
*/main

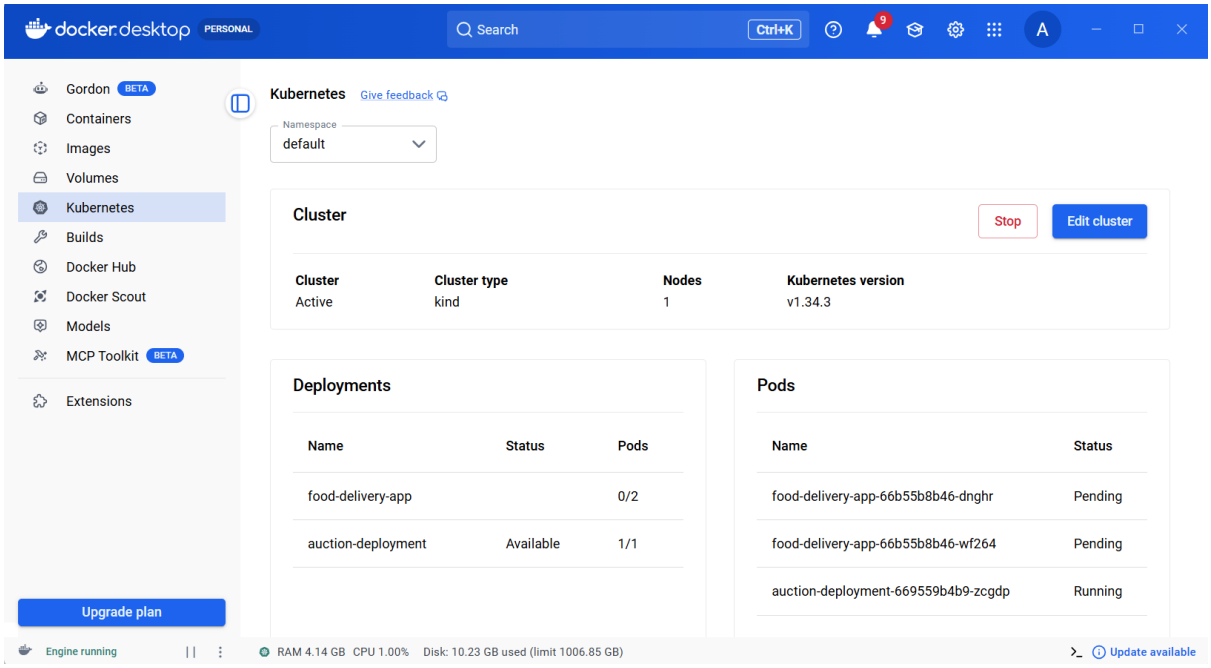
Jenkins / Onilne-auction-system / #3

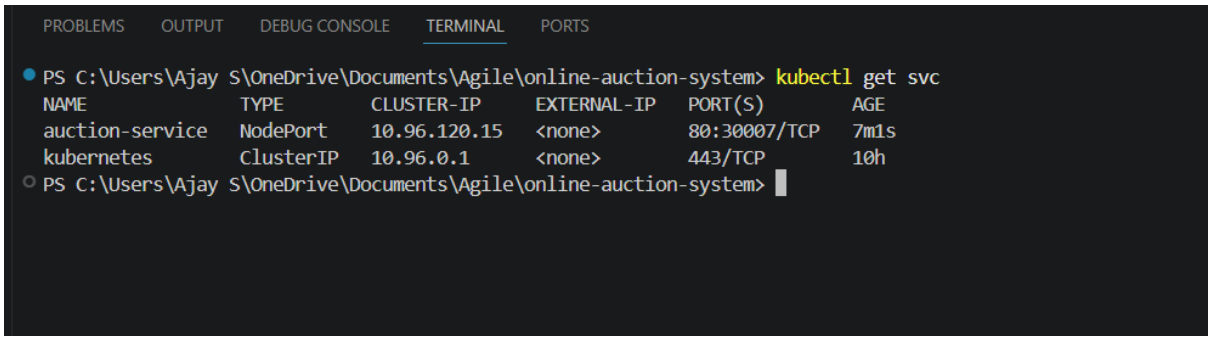
```
a149c623348d: Waiting
a149c623348d: Waiting
a149c623348d: Waiting
a149c623348d: Waiting
a149c623348d: Mounted from ajay222006/java-app
3fcc08d98cba: Pushed
0a470557608d: Pushed
latest: digest: sha256:df0a96487beaf15f5168e4bc30157e42a54b609fdb2aff1909ea1c551cbf0419 size: 856
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\Onilne-auction-system>kubectl apply -f k8s-deployment.yaml
deployment.apps/auction-deployment created
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\Onilne-auction-system>kubectl apply -f k8s-service.yaml
service/auction-service created
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```



iv)Deploy service on Docker and Kubernetes:-




docker desktop PERSONAL
Search
Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit BETA
Extensions
Upgrade plan
Kubernetes
Give feedback
Namespace
default
Cluster
Stop
Edit cluster
Cluster
Active
Cluster type
kind
Nodes
1
Kubernetes version
v1.34.3
Deployments
Name
Status
Pods
food-delivery-app
0/2
auction-deployment
Available
1/1
Pods
Name
Status
food-delivery-app-66b55b8b46-dnghr
Pending
food-delivery-app-66b55b8b46-wf264
Pending
auction-deployment-669559b4b9-zcgdp
Running
Engine running
RAM 4.14 GB CPU 1.00% Disk: 10.23 GB used (limit 1006.85 GB)
Update available


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system> kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
auction-service NodePort 10.96.120.15 <none> 80:30007/TCP 7m1s
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 10h
PS C:\Users\Ajay S\OneDrive\Documents\Agile\online-auction-system>